

**B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER
EXAM 2024**

Time: Three hours

Full Marks: 100

POWER SUPPLY SYSTEMS

(50 marks for each part)

Use separate answer script for each part.

PART I

Solutions

Question 1

Problem Statement: With suitable diagram discuss feed water and steam flow circuit.

Thought Process

Goal: Illustrate and explain the closed-loop water-to-steam-to-water cycle in a thermal plant.

Methodology: 1. Identify components: Boiler (Evaporator), Superheater, Turbine, Condenser, Feed pump, Economizer.

2. Draw a block diagram connecting these in a loop.

3. Explain the flow: Water is pumped to high pressure, preheated in the economizer, boiled in the drum, superheated, expanded in the turbine to produce work, condensed back to liquid in the condenser, and recirculated.

Answer:

Schematic Diagram:

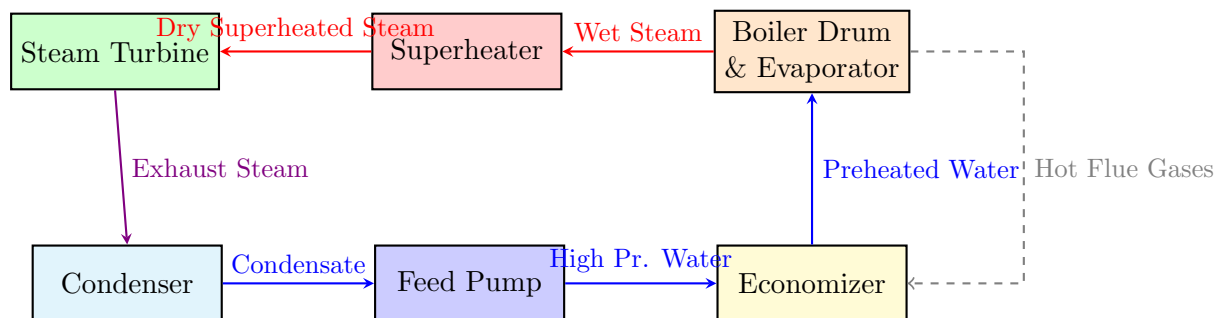


Figure 1: Feed Water and Steam Flow Circuit

Discussion of Flow Circuit:

The feed water and steam flow circuit is a closed thermodynamic loop based on the Rankine cycle.

1. **Condenser to Feed Pump:** The cycle begins at the condenser where exhaust steam from the turbine is cooled into liquid water (condensate). The Boiler Feed Pump (BFP) pressurizes this water to overcome boiler pressure.
2. **Economizer:** The high-pressure feed water passes through the economizer, where it extracts residual heat from outgoing flue gases, preheating the water and improving overall plant efficiency.
3. **Boiler:** The preheated water enters the boiler drum and circulates through water-wall tubes. The heat from fuel combustion converts the water into saturated (wet) steam.
4. **Superheater:** The wet steam is routed through the superheater, where it is heated by flue gases well above its saturation temperature. This creates dry, superheated steam, preventing moisture erosion in the turbine and raising thermal efficiency.
5. **Turbine to Condenser:** The superheated steam expands rapidly through the turbine blades, converting thermal energy into mechanical rotational energy (which drives the generator). The low-pressure, low-temperature exhaust steam drops into the condenser to be converted back into liquid, completing the loop.

Question 2

Problem Statement: Justify or correct the following sentences with reasons:

- a) Pumped storage plants are economic as peak load plant.
- b) Balanced draught must be used in thermal power plants.

Thought Process

Goal: Evaluate two technical statements and provide engineering justifications.

Methodology: a) Statement is correct. Explain the energy arbitrage (pumping during cheap off-peak hours, generating during expensive peak hours).

b) Statement requires slight correction/nuance. "Must" is strong, but it is highly preferred/used in modern large plants. Explain that it maintains a slightly negative pressure to prevent flame rollout, combining Forced Draught (FD) and Induced Draught (ID) fans.

Answer:

- a) **"Pumped storage plants are economic as peak load plant."**

Justification: Correct. Pumped storage plants act like massive, rechargeable batteries for the power grid. During off-peak hours (like nighttime), when electricity demand is low and generation from base-load plants (like coal or nuclear) is cheap and abundant, the plant uses this inexpensive power to pump water from a lower reservoir to an upper reservoir. During peak hours, when demand and electricity prices are at their highest, the stored water is released through turbines to generate high-value electricity. This energy arbitrage—buying cheap and selling high—makes them highly economical specifically for peak load management, despite consuming more total energy than they generate.

- b) **"Balanced draught must be used in thermal power plants."**

Justification: Correct (in the context of modern, large-capacity thermal plants). While small, older boilers might use only natural or forced draught, modern high-capacity thermal plants strictly use a balanced draught system.

Reasoning: A balanced draught system utilizes both a Forced Draught (FD) fan to push combustion air into the furnace and an Induced Draught (ID) fan to pull flue gases out. By

carefully synchronizing these fans, the pressure inside the furnace is maintained slightly below atmospheric pressure (a slight vacuum). If the pressure were positive (forced draught only), toxic flue gases and flames could leak out into the boiler house, posing a severe safety hazard. If the vacuum were too strong (induced draught only), excessive cold ambient air would leak into the furnace, drastically reducing thermal efficiency. Therefore, balanced draught is essential for safety, efficiency, and stable combustion.

Question 3(a)

Problem Statement: Derive an equation for air required for complete combustion of solid fuel considering the elements of that fuel C, H₂, O₂ and S.

Thought Process

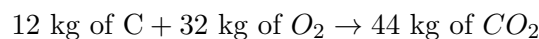
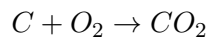
Goal: Formulate the stoichiometric air requirement formula based on fuel composition by weight.

Methodology: 1. Write the combustion equations for Carbon (C), Hydrogen (H₂), and Sulfur (S). 2. Calculate the mass of Oxygen (O₂) required for 1 kg of each element using atomic weights (C=12, O=16, H=1, S=32). 3. Sum the total oxygen required. Subtract the oxygen already present within the fuel. 4. Convert required oxygen mass to air mass, knowing air is $\approx 23\%$ oxygen by weight.

Answer:

Let C , H_2 , O_2 , and S be the mass fractions (in kg per kg of fuel) of Carbon, Hydrogen, Oxygen, and Sulfur respectively. We derive the oxygen required for each combustible element based on their chemical reactions. Atomic weights: $C = 12$, $H = 1$, $O = 16$, $S = 32$.

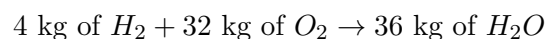
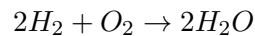
1. Combustion of Carbon:



$$\text{Therefore, 1 kg of } C \text{ requires } \rightarrow \frac{32}{12} = \frac{8}{3} \text{ kg of } O_2$$

For C kg of carbon, oxygen required = $\frac{8}{3}C$ kg.

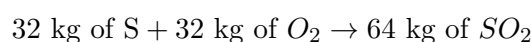
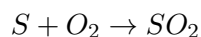
2. Combustion of Hydrogen:



$$\text{Therefore, 1 kg of } H_2 \text{ requires } \rightarrow \frac{32}{4} = 8 \text{ kg of } O_2$$

For H_2 kg of hydrogen, oxygen required = $8H_2$ kg.

3. Combustion of Sulfur:



$$\text{Therefore, 1 kg of } S \text{ requires } \rightarrow \frac{32}{32} = 1 \text{ kg of } O_2$$

For S kg of sulfur, oxygen required = S kg.

Total Oxygen and Air Requirement: Total theoretical oxygen required = $\left(\frac{8}{3}C + 8H_2 + S\right)$ kg. Since the fuel already contains O_2 kg of oxygen, the net oxygen to be supplied from outside is:

$$\text{Net } O_2 \text{ required} = \left[\frac{8}{3}C + 8H_2 + S - O_2\right] \text{ kg}$$

By weight, atmospheric air contains approximately 23% oxygen. Therefore, the mass of air required to supply 1 kg of oxygen is $\frac{100}{23}$ kg. The theoretical mass of air required per kg of fuel is:

$$\text{Mass of Air} = \frac{100}{23} \left[\frac{8}{3}C + 8\left(H_2 - \frac{O_2}{8}\right) + S\right] \text{ kg/kg of fuel} \quad (1)$$

Question 3(b)

Problem Statement: A thermal power plant spends Rs. 25 lakhs in one year as coal consumption. The coal has calorific value of 5000 kcal/kg and costs Rs. 500 per ton. If the thermal efficiency is 35% and electrical efficiency is 90%, find average load on the power plant.

Thought Process

Goal: Calculate the average electrical load (in kW or MW).

Methodology: 1. Find coal consumed per year: Cost / Rate. (Rs 2,500,000 / Rs 500/ton = 5000 tons = 5×10^6 kg). 2. Calculate total heat input per year: Mass $\times$ Calorific Value ($5 \times 10^6 \times 5000$ kcal). 3. Convert heat to electrical energy using efficiencies: Energy = Heat $\times \eta_{th} \times \eta_{elec}$. Convert kcal to kWh (1 kWh = 860 kcal). 4. Calculate average load: Total Energy / Total hours in a year (8760).

Answer:

Step 1: Calculate Total Coal Consumed Annually

Total annual cost = Rs. 25,00,000

Cost of coal = Rs. 500 per ton

Mass of coal consumed per year (M) = $\frac{25,00,000}{500} = 5000$ tons = 5×10^6 kg/year.

Step 2: Calculate Total Heat Input

Calorific Value (CV) = 5000 kcal/kg

Total Heat Input (H_{in}) = $M \times CV = (5 \times 10^6 \text{ kg}) \times (5000 \text{ kcal/kg})$

$H_{in} = 25 \times 10^9$ kcal/year.

Step 3: Convert Heat Input to Electrical Energy Output

Overall Efficiency ($\eta_{overall}$) = $\eta_{thermal} \times \eta_{electrical} = 0.35 \times 0.90 = 0.315$ (or 31.5%).

Useful Electrical Energy Out (E_{out}) = $H_{in} \times \eta_{overall} = 25 \times 10^9 \times 0.315 = 7.875 \times 10^9$ kcal/year.

To convert kcal to electrical units (kWh), we use the relation: 1 kWh = 860 kcal.

E_{out} in kWh = $\frac{7.875 \times 10^9}{860} = 9,156,976.74$ kWh/year.

Step 4: Calculate Average Load

Number of hours in a year = $24 \times 365 = 8760$ hours.

$$\begin{aligned} \text{Average Load} &= \frac{\text{Total Electrical Energy Output}}{\text{Total Hours}} \\ &= \frac{9,156,976.74 \text{ kWh}}{8760 \text{ h}} \\ &= 1045.31 \text{ kW} \end{aligned}$$

The average load on the power plant is 1045.31 kW (or ≈ 1.045 MW).

Question 4

Problem Statement: Discuss the functions of (i) dam (ii) penstock (iii) spillway (iv) valve house (v) tail race of a hydroelectric plant.

Thought Process

Goal: Define the civil/mechanical components of a hydro plant.

Methodology: State the primary structural/operational purpose of each element clearly and concisely.

Answer:

1. **Dam:** A massive barrier constructed across a river. Its primary function is to store a huge volume of water (creating a reservoir) and to artificially raise the water level to create a significant "head" (potential energy), which is necessary to drive the turbines.
2. **Penstock:** A large, reinforced conduit or pipe (usually made of steel or concrete) that carries water under high pressure from the reservoir or forebay downwards to the turbine casing.
3. **Spillway:** The safety valve of the dam. Its function is to safely discharge excess flood-water from the reservoir to the downstream side of the river without passing through the turbines, preventing the dam from overflowing and sustaining structural damage during heavy monsoons.
4. **Valve House:** A structure situated at the start of the penstock containing main sluice gates and automatic butterfly valves. Its function is to regulate the flow of water into the penstock and to completely shut off the water supply rapidly in case of emergencies or for maintenance of the turbines.
5. **Tail Race:** The channel or tunnel located at the discharge end of the turbine. Its function is to safely carry the low-energy water that has exited the turbine (after doing useful work) back into the natural river channel downstream.

Question 5

Problem Statement: Write short notes on: a) Starting motor of gas turbine plant, b) Fast breeder reactor.

Thought Process

Goal: Provide concise explanations of two specific power plant components.

Methodology: a) Explain that gas turbines are not self-starting; the motor cranks the compressor until ignition and self-sustaining speeds are reached.

b) Explain that an FBR uses fast neutrons (no moderator), produces more fissile material than it consumes (U-238 to Pu-239), and uses liquid sodium as a coolant.

Answer:

a) Starting Motor of Gas Turbine Plant:

Unlike a steam turbine, a gas turbine is not self-starting. For the Brayton cycle to initiate, the

main compressor must draw in and compress ambient air to a high pressure so that when fuel is injected and ignited, combustion can occur. A starting motor (which can be a large electric motor or a small diesel engine) is mechanically coupled to the turbine shaft. Its function is to crank the compressor to about 20% to 30% of its rated operational speed. At this speed, sufficient compressed air is generated, fuel is ignited, and the turbine begins to produce power. Once the turbine's generated power exceeds the power absorbed by the compressor (the self-sustaining speed), a clutch automatically disengages the starting motor.

b) Fast Breeder Reactor (FBR):

A Fast Breeder Reactor is an advanced nuclear reactor design that produces (breeds) more fissile material than it consumes.

- **No Moderator:** Unlike conventional reactors, FBRs do not use a moderator to slow down neutrons. They rely on highly energetic "fast neutrons" to sustain the fission chain reaction.
- **Breeding:** The core is surrounded by a "blanket" of fertile material, such as abundant Uranium-238. The fast neutrons escape the core and strike the U-238, transmuting it into Plutonium-239, a highly fissile fuel.
- **Coolant:** Because the core lacks a moderator, it is extremely compact and has a very high power density. Ordinary water cannot be used as a coolant because it acts as a moderator. Instead, liquid metals with excellent heat transfer properties, like liquid sodium, are used to cool the reactor.

Question 6

Problem Statement: Monthly flow data given. Draw flow duration curve and mass curve. Show average water flow. Calculate average hydro power (efficiency 85%, Head 100m). Calculate reservoir capacity if min stored volume is 500 month-sec-meter.

Thought Process

Goal: Plot hydrological curves, calculate power, and size a reservoir based on tabular data.

Data (m^3/s): J(400), F(200), M(600), A(1500), M(1000), J(2500), J(2300), A(2000), S(1200), O(1900), N(800), D(400).

Calculations: 1. Avg flow: Sum = 14800. Avg = $14800/12 = 1233.33 \text{ } m^3/s$. 2. Power: $P = \eta \rho g Q_{avg} H$. 3. Curves: FDC (sort descending), Mass Curve (cumulative). 4. Capacity: Find maximum consecutive cumulative deficiency from the average yield line, then add the minimum required dead storage (500).

Answer:

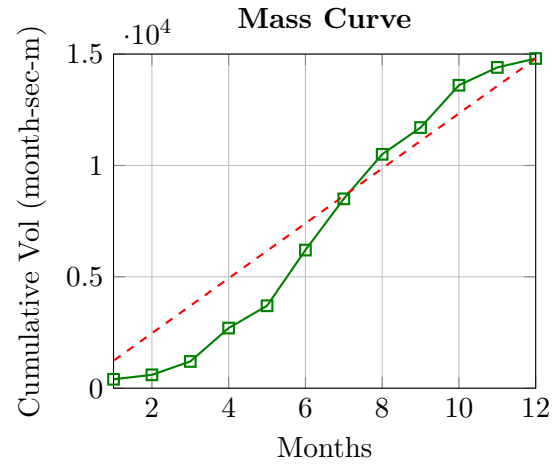
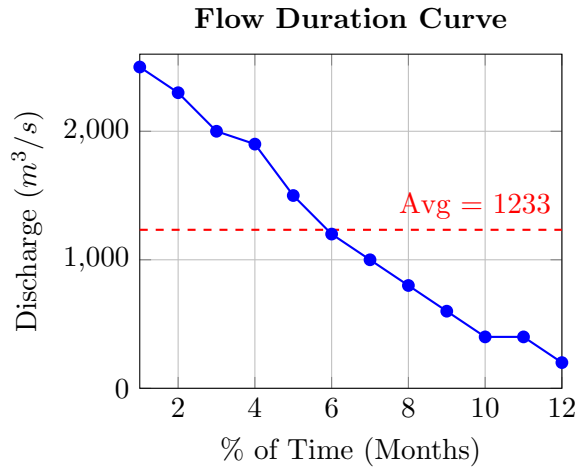
1. Data Preparation & Curves:

Total Flow = $400 + 200 + 600 + 1500 + 1000 + 2500 + 2300 + 2000 + 1200 + 1900 + 800 + 400 = 14800 \text{ } m^3/s$.

Average Flow (Q_{avg}) = $\frac{14800}{12} = 1233.33 \text{ } m^3/s$.

Flow Duration Data (Sorted Descending): 2500, 2300, 2000, 1900, 1500, 1200, 1000, 800, 600, 400, 400, 200.

Mass Curve Data (Cumulative Flow): Jan(400), Feb(600), Mar(1200), Apr(2700), May(3700), Jun(6200), Jul(8500), Aug(10500), Sep(11700), Oct(13600), Nov(14400), Dec(14800).



2. Calculate Average Hydro Power:

Using the formula $P = \eta \cdot \rho \cdot g \cdot Q_{avg} \cdot H$

$$\eta = 0.85$$

$$\rho = 1000 \text{ kg}/m^3 \text{ (density of water)}$$

$$g = 9.81 \text{ m}/s^2$$

$$Q_{avg} = 1233.33 \text{ m}^3/s$$

$$H = 100 \text{ m}$$

$$P = 0.85 \times 1000 \times 9.81 \times 1233.33 \times 100 = 1,028,401,395 \text{ Watts} = \mathbf{1028.4 \text{ MW}}.$$

3. Calculate Reservoir Capacity:

The required active capacity is the maximum cumulative deficit from the average yield. Let's calculate the monthly deficit (Inflow - Demand) where Demand is $Q_{avg} = 1233.33$. Assuming the reservoir is full at the end of October (wet season):

- Nov: $800 - 1233.33 = -433.33$
- Dec: $400 - 1233.33 = -833.33$
- Jan: $400 - 1233.33 = -833.33$
- Feb: $200 - 1233.33 = -1033.33$
- Mar: $600 - 1233.33 = -633.33$
- Apr: $1500 - 1233.33 = +266.67$ (Start of filling)

The maximum consecutive accumulated deficit is from Nov to Mar: Max Deficit = $433.33 + 833.33 + 833.33 + 1033.33 + 633.33 = \mathbf{3766.65}$ month-sec-meter. Total Reservoir Capacity = Active Storage (Max Deficit) + Dead Storage (Minimum Volume) Total Capacity = $3766.65 + 500 = \mathbf{4266.65}$ month-sec-meter.

Question 7

Problem Statement: a) Distinguish between load curve and load duration curve. b) Deduce expression of optimum operating time of peak load plant.

Thought Process

Goal: Define load curves and derive the optimal operating hours formula.

Methodology: a) Compare axes and chronological vs sorted nature. b) Formulate cost equations for Base (C_b) and Peak (C_p) plants in the form $C = A \cdot \text{kW} + B \cdot \text{kWh}$. Equate the cost of generating 1 kW for H hours to find the break-even point.

Answer:

a) Distinguishing between Load Curve and Load Duration Curve:

Load Curve	Load Duration Curve (LDC)
Plots electrical load (MW) against the chronological time of day (e.g., 0 to 24 hrs).	Plots electrical load (MW) arranged in descending order of magnitude against duration.
Shows exactly *when* the peak loads and base loads occur during the day.	Shows *how long* a certain load magnitude persists, but loses the specific time of occurrence.
Used heavily by dispatchers to anticipate daily grid operations and turn units on/off.	Used for economic analysis, calculating load factors, and sizing base/peak plant capacities.

b) Optimum Operating Time of a Peak Load Plant:

Let us define the annual operating cost function for a power plant, which consists of fixed costs (dependent on installed capacity, kW) and variable running costs (dependent on energy generated, kWh). Let the plant operate for t hours per year.

For the **Base Load Plant** (e.g., Steam): $C_b = A_b \times (\text{Capacity in kW}) + B_b \times (\text{Energy in kWh})$
To generate 1 kW of capacity for t hours, the annual cost is:

$$C_b = A_b + B_b \cdot t \quad (2)$$

Where A_b is high (high capital cost) and B_b is low (cheap fuel).

For the **Peak Load Plant** (e.g., Gas Turbine): $C_p = A_p \times (\text{Capacity in kW}) + B_p \times (\text{Energy in kWh})$

To generate 1 kW of capacity for t hours, the annual cost is:

$$C_p = A_p + B_p \cdot t \quad (3)$$

Where A_p is low (low capital cost) and B_p is high (expensive fuel).

Deduction: To find the optimum operating hours (H) for the peak load plant, we must find the break-even point where the cost of generating power from the peak load plant exactly equals the cost from the base load plant. If a load lasts longer than H , it is cheaper to supply it with the base plant. If it lasts less than H , it is cheaper to use the peak plant.

Equating the two cost functions at $t = H$:

$$\begin{aligned} C_b &= C_p \\ A_b + B_b \cdot H &= A_p + B_p \cdot H \\ A_b - A_p &= H(B_p - B_b) \end{aligned}$$

Therefore, the optimum operating time (in hours/year) of the peak load plant is:

$$H = \frac{A_b - A_p}{B_p - B_b} \quad (4)$$